

Donghwan Chi

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Interests

Artificial General Intelligence, Multimodal Large Language Models, Vision-Language-Speech Generation, Object-Centric Representation Learning, Text-to-Image Generation, Multimodal Alignment

Education

- M.S. Korea University**, Artificial Intelligence March 2025 – Present
- GPA: 4.44/4.5
- B.S. Korea University**, Computer Science and Engineering March 2019 – Feb. 2025
- GPA: 4.11/4.5
 - Leave of absence for mandatory military service in the Republic of Korea Air Force, Jan. 2021 – Jan. 2023

Publications

- Slot-MLLM: Object-Centric Visual Tokenization for Multimodal LLM** Submitted to IEEE Transactions on Multimedia.
- Donghwan Chi***, Hyomin Kim*, Yoonjin Oh, Yongjin Kim, Donghoon Lee, Daejin Jo, Jongmin Kim, Junyeob Baek, Sungjin Ahn, Sungwoong Kim * *Equal contribution.*
- Object-centric Self-improving Preference Optimization for Text-to-Image Generation** Accepted to CVPR 2026.
- Yoonjin Oh, Yongjin Kim, Hyomin Kim, **Donghwan Chi**, Sungwoong Kim

Research Experience

- WAVLab**, Carnegie Mellon University Pittsburgh, USA (Online)
- Visiting Collaborator (Advisor: Prof. Shinji Watanabe)* Oct. 2025
- Conducted research on vision-speech-language omni-modal models for unified multi-modal understanding and generation.
 - Developed alignment-preserving training methods to extend large vision-language models with discrete speech tokenizers for ITS2ITS omni-modal understanding and generation across image, text, and speech.
- AGI Lab**, Korea University Seoul, South Korea
- Graduate Researcher / Undergraduate Intern (Advisor: Prof. Sungwoong Kim and Prof. Sungbin Lim)* Sept. 2023
- Led research on object-centric visual tokenization for multimodal large language models, resulting in Slot-MLLM.
 - Designed a Slot Q-Former-based visual tokenizer to improve object-level grounding, generation, and editing.
 - Conducted large-scale multimodal training on image-text and interleaved datasets using multi-GPU distributed training.
 - Evaluated object-centric reasoning and generation on benchmarks including spatial relation understanding, text-to-image generation, and image editing.
 - Managed lab GPU servers and supported large-scale training infrastructure.

Extracurricular Activities

Artificial Intelligence Korea University (AIKU),

First Cohort Senior Member, Technical & Academic Management Team

Seoul, South Korea

Jan. 2023 – Jan. 2024

1 year

- Managed the club's servers, ensuring reliable and efficient operation.
- Served as a committee member for multiple hackathon question designs.
- Conducted and participated in a project related to diffusion models.
- Organized and taught introductory courses for new members.

Honors and Awards

- **Chief of Naval Operations Republic of Korea Navy Award**, Open Source Academy Military (OSAM) Hackathon, 2021
- **Semester High Honors**, Korea University, 2019, 2020, 2023, 2024
- **Special Scholarship by Student Affairs Office**, Korea University, Fall 2023, Spring 2024
- **Academic Excellence Scholarship**, Korea University, Spring 2024
- **Excellence Award**, Capstone Design Competition, Korea University, Fall 2024

Technologies

Languages: Python, Bash, LaTeX

ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers

ML Systems: Distributed Training, Multi-GPU Training, Mixed Precision Training, DeepSpeed, Weights & Biases

Research Areas: Multimodal LLMs, Vision-Language Models, Diffusion Models, Speech Tokenization

Language Proficiency: TOEIC 905